INCH-POUND
MIL-DTL-18/91F
6 October 2023
SUPERSEDING
MIL-B-18/91E
W/AMENDMENT 1
7 July 2021

MILITARY SPECIFICATION SHEET

BATTERY, DRY, BA-413/U

INACTIVE FOR NEW DESIGN AFTER
10 AUGUST 1998 AND IS NO LONGER USED,
EXCEPT FOR REPLACEMENT PURPOSES.

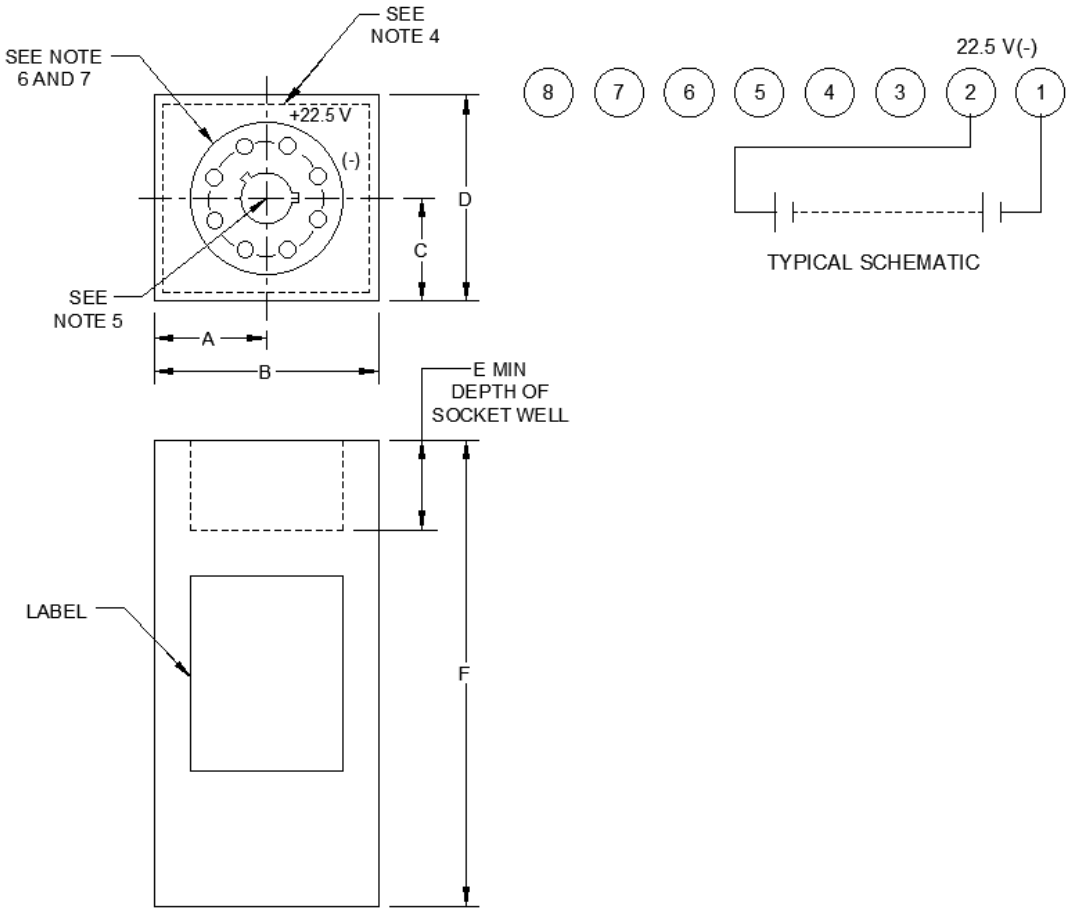
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-18.

AMSC N/A
FSC 6135

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Ltr	Inches Min	Inches Max	mm Min	mm Max
A	.718	.844	18.24	21.44
B	1.500	1.626	38.10	41.30
C	.655	.781	16.64	19.84
D	1.375	1.501	34.93	38.13
E	.562	.688	14.27	17.48
F	3.187	3.313	80.95	84.15

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.063 inch (1.600 mm).
4. Size of insulating plate at base of socket is optional.
5. Pin-circle center.
6. The jacket opening shall be 1.063 inch (27.00 mm) minimum diameter and shall be positioned so that its center is within a .063 inch (1.588 mm) diameter circle whose center coincides with the pin-circle center.
7. Socket type B is required. (See MIL-B-18 for details.)
8. Marking to be in accordance with MIL-B-18.

FIGURE 1. Battery dimensions.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Nominal voltage: 2.25 volts.

Usual number and type of cells: 15 “N” cells.

Usual cell connection: Series.

Terminals: 8-hole socket.

Weight (maximum): 6 ounces.

Capacity tests: When the battery is tested in accordance with the methods of examination and test of this specification, the minimum capacity-test requirements shall be not less than the minimum time specified for SLD or SLT.

First article inspection in accordance with MIL-DTL-18:

Visual and mechanical (external).
Battery voltage.
Vibration test.
Mechanical-shock test in accordance with MIL-STD-202-213, test condition I.
Insulation resistance test in accordance with MIL-STD-202-302, test condition B, tolerance ±20 volts.
Capacity, D (without storage).
Jacket integrity test.

Conformance inspection in accordance with MIL-DTL-18:

Visual and mechanical (external).
Battery voltage.
Insulation resistance test in accordance with MIL-STD-202-302, test condition B, tolerance ±20 volts.
Vibration test.
Mechanical-shock test in accordance with MIL-STD-202-213, test condition I.
Jacket integrity test.
Electrode leakage.
Capacity, D and T.

Method of examination and test:

Capacity tests: See requirements for capacity specified herein.

Storage:

Test	Period
D	12 months
T	90 days

Discharge: The battery shall be discharged through 1,300 ohms for 2 minutes, then through 3,300 ohms for 18 minutes for a period of 4 hours, then on open circuit for 20 hours. This cycle shall be repeated 5 days per week continuously to a test-end voltage of 16.5 volts.

Closed circuit voltage: Use TS-183 ( )/U test per special marking or, with minimum permissible

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voltage as specified for TS-183 ( )/U, use a load resistance value of 845.4 ohms.

Special marking on each unit package:

TO TEST THIS BATTERY WITH TS-183 ( )/U:

USE JACK NO.	MINIMUM PERMISSIBLE VOLTAGE
14	20.7

Unless expressly authorized, this test information shall apply to Army and Air Force applications only.

Part or Identifying Number (PIN): BA-413/U, see 1.2 of MIL-DTL-18.

Change notations. The margins of this standard are marked with vertical lines to indicate modifications generated by this change. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-18, this document references the following:

MIL-STD-202-213 MIL-STD-202-302

Custodians:
Army - CR
Navy - SH
DLA - CC

Review activities:
Army - MI
Navy - MC

Preparing activity:
DLA - CC

(Project 6135-2023-010)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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